# Evaluating and Training Large Language Models: What Can (and Cannot) Item Parameters Tell Us?

Research Group Seminar · October 26

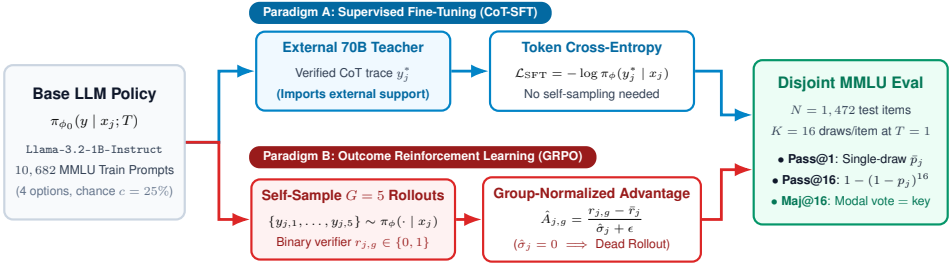
## Standard LLM Practice

- **Evaluation**: Rank models by average single-draw accuracy ($\mathrm{Pass@1}$) or a single 1D ability score $\hat{\theta}$.
- **Training**: Plug static population item parameters $(a_j, b_j)$ into RL advantage estimation.
- **Limitation**: Aggregate averages hide *which* items change and *why*.

## What an IRT Lens Reveals

- **Where Static IRT Fails**: Population $(a_j, b_j)$ reflects cross-model scale (memorization vs. math), not within-model learning ($\mathrm{Pass@1}$ gain $= 0$).
- **Where IRT Succeeds**: Modeling a single LLM across repeated trials separates **curve steepening ($\tau \leftrightarrow 1/T$)** from **knowledge shift ($\Delta\theta$)**.

# Background: LLM Post-Training & Evaluation

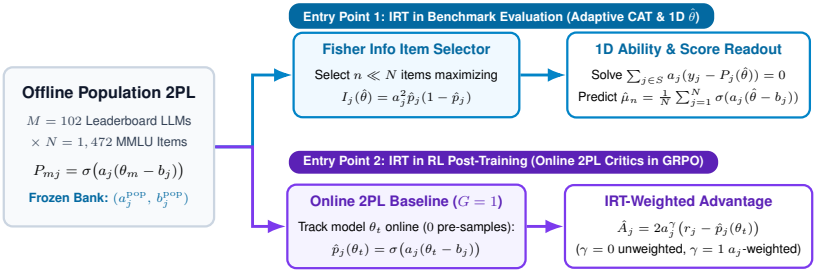



**CoT-SFT (Supervised)**
Trains on verified 70B teacher traces $y_j^*$.
→ **Imports external knowledge** even when Base pass rate $p_j = 0$.

**GRPO (Outcome RL)**
Samples $G = 5$ self-rollouts; rewards binary correctness $r_{j,g} \in \{0, 1\}$.
→ **Zero gradient** if all $G$ rollouts fail or succeed ($\hat{\sigma}_j = 0$).

**Test Metrics ($K = 16$ draws)**
- $\mathbf{Pass@1}$: Mean single-draw accuracy.
- $\mathbf{Pass@16}$: At least 1 correct in 16 draws.
- $\mathbf{Maj@16}$: Majority-vote accuracy.

# Background: How IRT Enters LLM Pipelines




**Entry 1: Adaptive Evaluation (CAT)**
Calibrate 2PL $(a_j, b_j)$ on $M = 102$ leaderboard LLMs (Maia Polo et al., 2024; Kipnis et al., 2025).
$\rightarrow$ Select high-information items $I_j(\hat{\theta}) = a_j^2 \hat{p}_j(1 - \hat{p}_j)$ to estimate score from $n \ll N$ items.

**Entry 2: Online 2PL Critics in RL**
Replace $G = 5$ group rollouts with a single rollout ($G = 1$) using IRT expectation $\hat{p}_j(\theta_t) = \sigma(a_j(\theta_t - b_j))$:

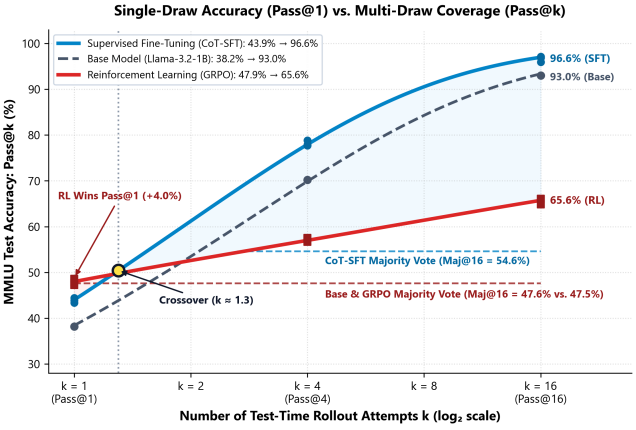
$$\hat{A}_j = 2\, a_j^{\gamma}\big(r_j - \hat{p}_j(\theta_t)\big), \qquad \gamma \in \{0, 1\}$$

# Terminology Bridge: LLM vs. Psychometric IRT

| Concept | LLM / ML Term | Psychometric / IRT Term | Notation |
|---|---|---|---|
| 1. Test-Taker | Model checkpoint $\pi_\phi$ ($K = 16$ draws) | Respondent with ability $\theta$ | $\pi_\phi \leftrightarrow \theta$ |
| 2. Item Easiness | Base initial pass rate ($p_{\text{base},j}$) | Within-model item intercept ($\mu_{0j}$) | $\mu_{0j} = \text{logit}(p_{\text{base},j})$ |
| 3. Sharpening | Inverse temperature ($1/T$) or RL | Response curve slope ($\tau$) | $\tau \leftrightarrow 1/T$ |
| 4. Chance Floor | 4-choice guessing rate ($25\%$) | 3PL guessing parameter ($c = 1/K$) | $c = 0.25$ |
| 5. Learning Signal | `GRPO` reward variance ($\hat{\sigma}_j^2 > 0$) | 1PL item Fisher information | $I_j = p_j(1 - p_j)$ |
| 6. Prompt Bias | Option-order gap (`A/B/C/D` swap) | Differential Item Functioning (DIF) | $D = \lvert p_{\text{ord0}} - p_{\text{ord1}} \rvert$ |

**Key Distinction:** **Population IRT** calibrates $(a_j, b_j)$ across 102 *different* models (between-model). **Within-model IRT** tracks how a *single* model's pass probability $p_j$ evolves across repeated rollouts during training.

# The Empirical Puzzle: Pass@1 vs. Pass@k




**1. Single-Draw** $\mathrm{Pass@1}$ **(**$k = 1$**)**

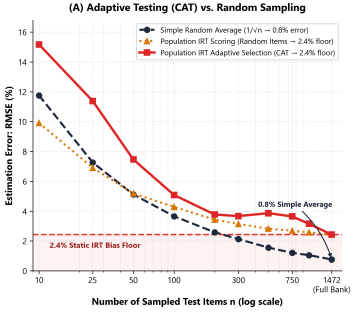
- `Base`: $\mathbf{38.2}\%$
- `CoT-SFT`: $\mathbf{43.9}\%$ $(+5.7\%)$
- `GRPO (RL)`: $\mathbf{47.9}\%$ $(+9.7\%$, wins at $k = 1)$

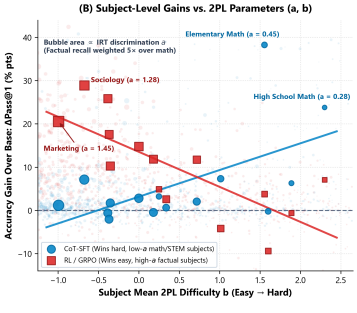
**2. Multi-Draw** $\mathrm{Pass@16}$ **&** $\mathrm{Maj@16}$

- $\mathrm{Pass@16}$ **collapses under RL**: `GRPO` drops to $\mathbf{65.6}\%$ vs. `Base` $\mathbf{93.0}\%$ and `CoT-SFT` $\mathbf{96.6}\%$.
- **Majority Vote (**$\mathrm{Maj@16}$**) is unchanged by RL**: `GRPO` $(\mathbf{47.5}\%) =$ `Base` $(\mathbf{47.6}\%)$, whereas `CoT-SFT` reaches $\mathbf{54.6}\%$.

$\rightarrow$ **Intuition**: `GRPO` learns **0% net new modal answers**; it only concentrates probability onto existing Base modes.

# IRT in Evaluation: Adaptive Testing & Domain Bias



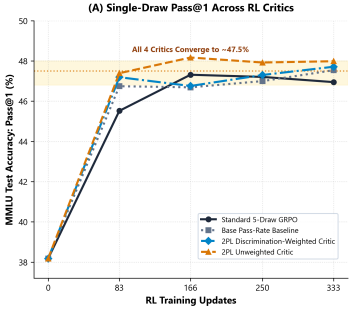

**(A) Adaptive CAT Hits a $2.4\%$ Error Floor**

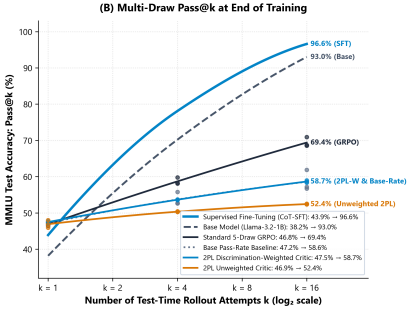
- Simple random sampling reaches $\mathbf{0.8}\%$ RMSE at full bank.
- 1D IRT scoring & CAT plateau at $\mathbf{2.4}\%$ error because post-trained models violate 1D invariance.

**(B) $5\times$ Discrimination Skew Across Subjects**

- Factual recall has $5\times$ higher $a_j$ (*Marketing* $\mathbf{1.45}$) than math (*HS Math* $\mathbf{0.28}$). Since $I_j \propto a_j^2$, CAT over-samples memory $25\times$.
- GRPO gains on easy/high-$a$ items; CoT-SFT gains on hard/low-$a$ math.

# IRT in RL Training: Online 2PL Critics



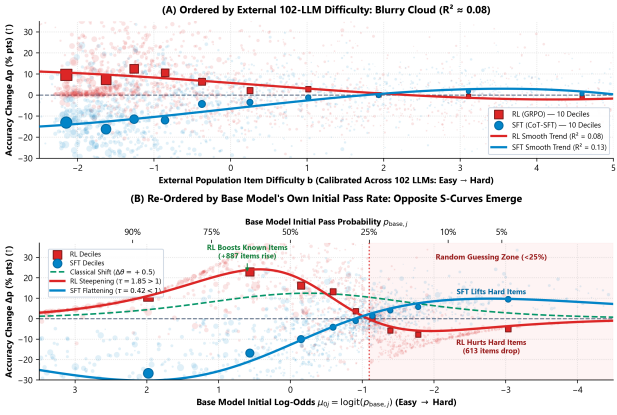

**(A) All 4 Critics Saturate at $\sim 47.5\%$ Pass@1**

- Standard `GRPO` ($G = 5$) and single-rollout ($G = 1$) 2PL critics all converge to $\mathbf{46.8\%}$–$\mathbf{47.5\%}$.
- Online 2PL baseline cuts sampling cost $5\times$, but $a_j$-weighting adds **zero accuracy gain**.

**(B) High-$k$ Collapse & Item-Subset Divergence**

- At $k = 16$, all RL critics collapse ($\mathbf{52.4\%}$–$\mathbf{69.4\%}$) far below `Base` ($\mathbf{93.0\%}$).
- Despite tied aggregate Pass@1, **5 of 6 critic pairs** learn statistically distinct item subsets ($p < 0.05$).

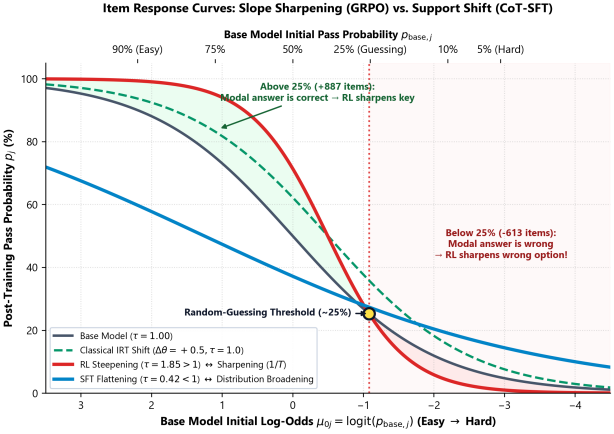
# Why Population Parameters Fail in Training




**1. Wrong Coordinate Frame (Panel A)**
Ordering item gains $\Delta p_j$ by 102-LLM difficulty $b_j$ gives a flat cloud ($\mathbf{R^2 = 0.08}$).
$\rightarrow$ Policy updates are nearly orthogonal to population $a_j$ ($\mathbf{r = 0.072}$).

**2. Base Model Pass Rate Frame (Panel B)**
Ordering by the Base model's own initial log-odds $\mu_{0j} = \text{logit}(p_{\text{base},j})$ reveals two opposite S-curves ($\mathbf{R^2 = 0.35}$):
- **Above** $25\%$ **chance**: `GRPO` boosts $+887$ already-known items.
- **Below** $25\%$ **chance**: `GRPO` hurts $613$ hard items, whereas `CoT-SFT` lifts them.

# Mechanism: Curve Steepening vs. Support Shift




## 1. Within-Model Response Curve

Post-training pass rate as a function of initial log-odds $\mu_{0j}$:

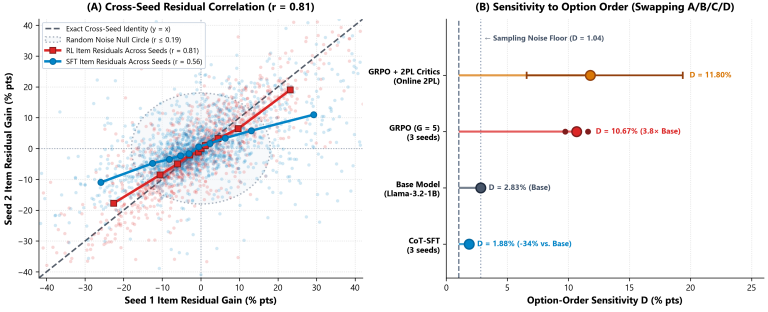
$$p_j = \sigma\big(\alpha + \tau(\mu_{0j} - \mu_c)\big), \quad \mu_c = \text{logit}(0.25)$$

- **Slope** $\tau \leftrightarrow 1/T$: Sharpens existing preferences.
- **Shift** $\alpha$: Adds new knowledge support.

## 2. Why $c = 25\%$ Is the Polarity Pivot

- GRPO steepens the slope ($\tau = \mathbf{1.85} > \mathbf{1}$):
  – When $p_{\text{base},j} > 25\%$, the modal answer is correct → sharpening increases accuracy.
  – When $p_{\text{base},j} < 25\%$, a **wrong distractor is modal** → sharpening locks in the wrong answer ($p_j \to 0$).
- CoT-SFT ($\tau = \mathbf{0.42} < \mathbf{1}$) shifts support upward on hard items.

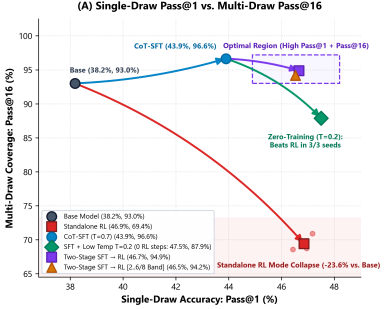
# Diagnostics: Cross-Seed Residuals & Option Order

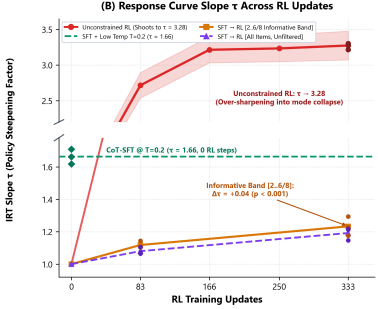



**(A) Cross-Seed Residual Correlation ($\mathbf{r = 0.81}$)**
After removing the 1D curve, item residuals across random seeds remain highly correlated ($\mathbf{r = 0.81}$ in RL vs. $0.19$ noise floor).
$\rightarrow$ Item-level learning is deterministic and multi-dimensional, set by Base initialization.

**(B) RL Amplifies Option-Order Bias $4\times$**
Swapping option letters (`A/B/C/D`) measures position sensitivity $D$:
- `Base`: $\mathbf{2.83\%}$ | `CoT-SFT`: $\mathbf{1.88\%}$ ($-34\%$ bias)
- `GRPO`: $\mathbf{10.67\%}$ ($4\times$ **position bias** when sharpening low-confidence items).

# Implications: Decoding Temperature & Informative Rollouts




**1. Temperature Shortcut**
Since RL acts like $\tau \leftrightarrow 1/T$, decoding `CoT-SFT` at $\mathbf{T = 0.2}$ (**0 RL steps**) jumps to $\mathbf{47.5}\%$ Pass@1 while keeping $\mathbf{87.9}\%$ Pass@16.

**2. Two-Stage SFT → RL**
Shifting support first via `CoT-SFT` and then running RL achieves both high Pass@1 ($\mathbf{46.7}\%$) and Pass@16 ($\mathbf{94.9}\%$).

**3. Informative Band $[2..6/8]$**
Filtering items with variance $p_j(1-p_j) > 0$ cuts dead rollouts $\mathbf{34}\% \rightarrow \mathbf{23}\%$ and steepens $\tau$ ($+\mathbf{0.04}, \mathbf{p < 0.001}$).

# Summary: What IRT Reveals Beyond Aggregate Scores

| Question | Standard ML View | IRT / Psychometric Concept | Core Finding |
|---|---|---|---|
| **1. Benchmark Eval** | 1D score $\hat{\theta}$ from $32$–$100$ adaptive items suffices. | **2PL Factor Loading $(a_j, b_j)$ & Invariance** | $a_j$ is $\mathbf{5}\times$ higher on factual memory than math $\rightarrow$ $\mathbf{2.4}\%$ CAT bias floor. |
| **2. RL Critics** | Weighting RL advantages by $a_j$ should boost accuracy. | **Between- vs. Within-Model Coordinate Frame** | All 4 critics tie at $\sim \mathbf{47.5}\%$ $\mathrm{Pass@1}$ $(\mathrm{corr}(J_j u_\perp, a_j) = 0.072)$. |
| **3. SFT vs. RL** | RL improves reasoning $(\mathrm{Pass@1} : 38.2\% \rightarrow 47.9\%)$. | **ICC Slope ($\tau \leftrightarrow 1/T$) & $c = 25\%$ Guessing Threshold** | RL learns **0% new modes** $(\mathrm{Maj@16} = 47.5\%)$; sharpens wrong distractors below $25\%$. |
| **4. Stability** | Ignore item residuals and option order. | **Local Independence & Format Sensitivity ($D$)** | Cross-seed residuals correlate at $\mathbf{r = 0.81}$; RL amplifies order bias $\mathbf{4}\times$ $(D = 10.67\%)$. |
| **5. Control** | Tune RL via aggregate validation $\mathrm{Pass@1}$. | **Item Variance $p_j(1-p_j)$ & Slope Tracker $\tau_t$** | CoT-SFT @ $T = 0.2$ hits $\mathbf{47.5}\%$ with 0 RL steps; $[2..6/8]$ band cuts dead rollouts $\mathbf{34}\% \rightarrow \mathbf{23}\%$. |

# Next Steps

### 1. Domain-Balanced CAT

Constrain adaptive item selection across memory vs. reasoning dimensions so high-$a_j$ factual items do not dominate evaluation.

### 2. Distractor-Aware RL

Model all 4 option probabilities $(p_A, p_B, p_C, p_D)$ to prevent confidence sharpening on wrong modal distractors when $p_{\text{base},j} < 25\%$.

### 3. Trajectory Calibration

Exploit high cross-seed repeatability ($r = 0.81$) to calibrate item parameters along training checkpoints rather than static leaderboards.

**Thank You!**

# Backup B1: First-Order Expansion & Full Results

## First-Order Logit Expansion

For policy log-odds $z_j(\phi) = \operatorname{logit} p_j(\phi)$ under parameter update $\Delta\phi = \alpha_\parallel u_{\text{pop}} + u_\perp$:

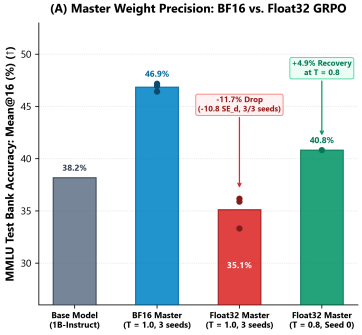
$$\Delta p_j \approx p_{0j}(1 - p_{0j})\left[\alpha_\parallel a_j^{\text{pop}} + J_j u_\perp\right]$$

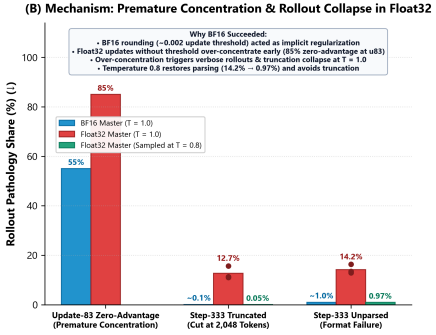
- $\operatorname{corr}(J_j u_\perp, a_j^{\text{pop}}) = \mathbf{0.072}$.
- Within-model variance $p_{0j}(1 - p_{0j})$ drives $\Delta p_j$ ($R^2 = 0.354$), while $a_j^{\text{pop}}$ has near-zero alignment ($r = +0.025$).

## Disjoint MMLU Test Bank ($N = 1,472$)

| Model / Arm | Draws | Pass@1 | Pass@16 | Maj@16 | Order $D$ |
|---|---|---|---|---|---|
| Base (Llama-3.2-1B) | — | 38.2% | 93.0% | 47.6% | 2.83% |
| CoT-SFT (70B Teacher) | 0 | 43.9% | **96.6%** | **54.6%** | **1.88%** |
| GRPO Standard ($G = 5$) | 5 | 46.9% | 69.4% | 47.5% | 10.67% |
| Base Pass-Rate Baseline | 1 | 47.2% | 58.6% | 47.4% | 11.42% |
| 2PL Unweighted ($\gamma = 0$) | 1 | 46.9% | 52.4% | 47.1% | 12.15% |
| 2PL $a_j$-Weighted ($\gamma = 1$) | 1 | **47.5%** | 58.7% | 47.5% | 11.80% |
| CoT-SFT @ T=0.2 (0 RL) | 0 | **47.5%** | 87.9% | 54.6% | 1.95% |
| Two-Stage SFT → RL | 1 | 46.7% | 94.9% | 54.4% | 3.12% |
| Two-Stage [2..6/8 Band] | 1 | 46.5% | 94.2% | 54.5% | 2.98% |

# Backup B2: Numerical Precision Audit (BF16 vs. Float32)




**(A) Float32 Master Weights Drop Accuracy by $-11.7\%$**

Removing BF16 rounding ($\sim 0.002$ threshold) lowers $\mathrm{Pass@1}$ from $\mathbf{46.9}\% \rightarrow \mathbf{35.1}\%$ (partially recovered at $T = 0.8$).

**(B) Premature Concentration in Float32**

Exact Float32 updates over-concentrate early ($\mathbf{85}\%$ zero-variance rollouts vs. $55\%$ in BF16), causing truncation ($12.7\%$) and parse errors ($14.2\%$).